Exp. No: 4

Measurement and Analysis of Dispersion Compensation in Optical Fiber using OptiSystem

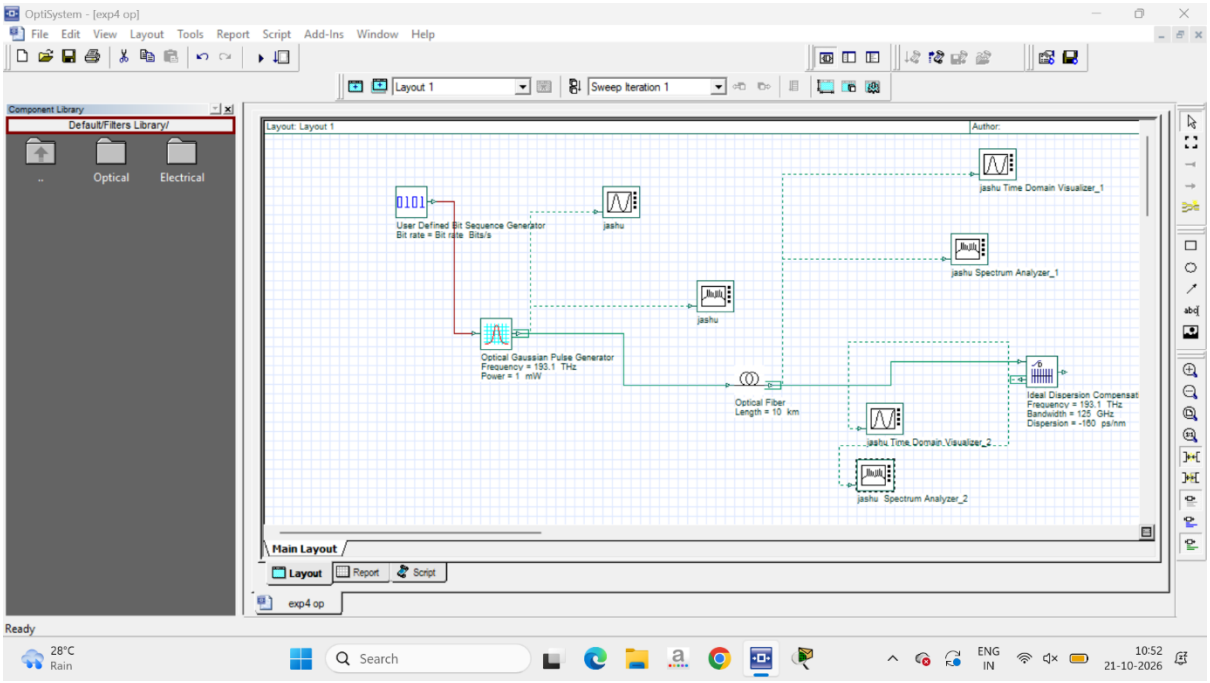
OBJECTIVES

1. To demonstrate dispersion compensation using the Ideal Dispersion Compensation component in OptiSystem.
2. To observe the broadening of a 12.5 ps optical Gaussian pulse after propagation through 10 km of single-mode fiber (SMF).
3. To compensate the accumulated dispersion of 160 ps/nm and verify recovery of the original pulse shape.

SOFTWARE REQUIRED

OptiSystem Software

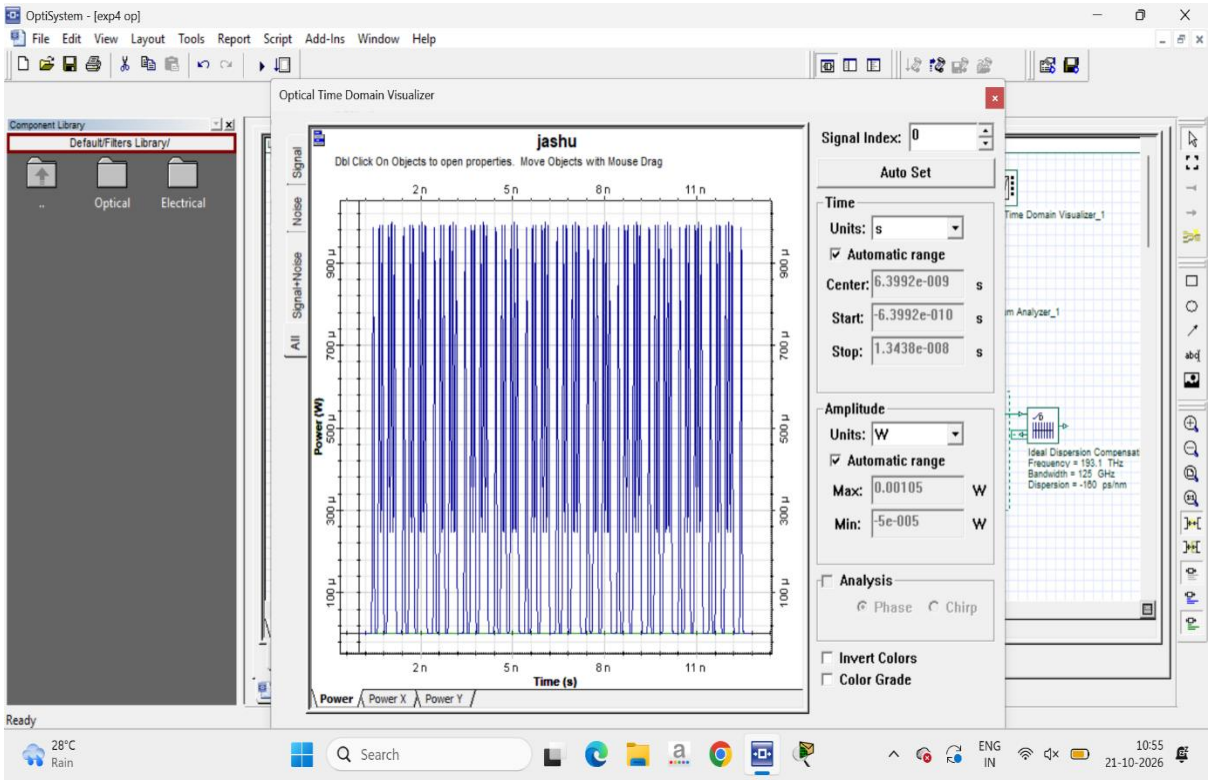
CIRCUT DIAGRAM:



OUTPUT GRAPH:

Figure 1: Initial Gaussian pulse

A.


OptiSystem - [exp4 op]
File Edit View Layout Tools Report Script Add-Ins Window Help
Optical Time Domain Visualizer
jashu
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Signal Index: 0
Auto Set
Time
Units: s
Automatic range
Center: 6.3992e-009 s
Start: -6.3992e-010 s
Stop: 1.3438e-008 s
Amplitude
Units: W
Automatic range
Max: 0.00105 W
Min: -5e-005 W
Analysis
Phase Chirp
Invert Colors
Color Grade
Time (s)
Power (W)
Power Power X Power Y
Component Library
Default/Filters Library/
Optical
Electrical
Ready
28°C Rain
Search
ENG IN
10:55 21-10-2026

B.

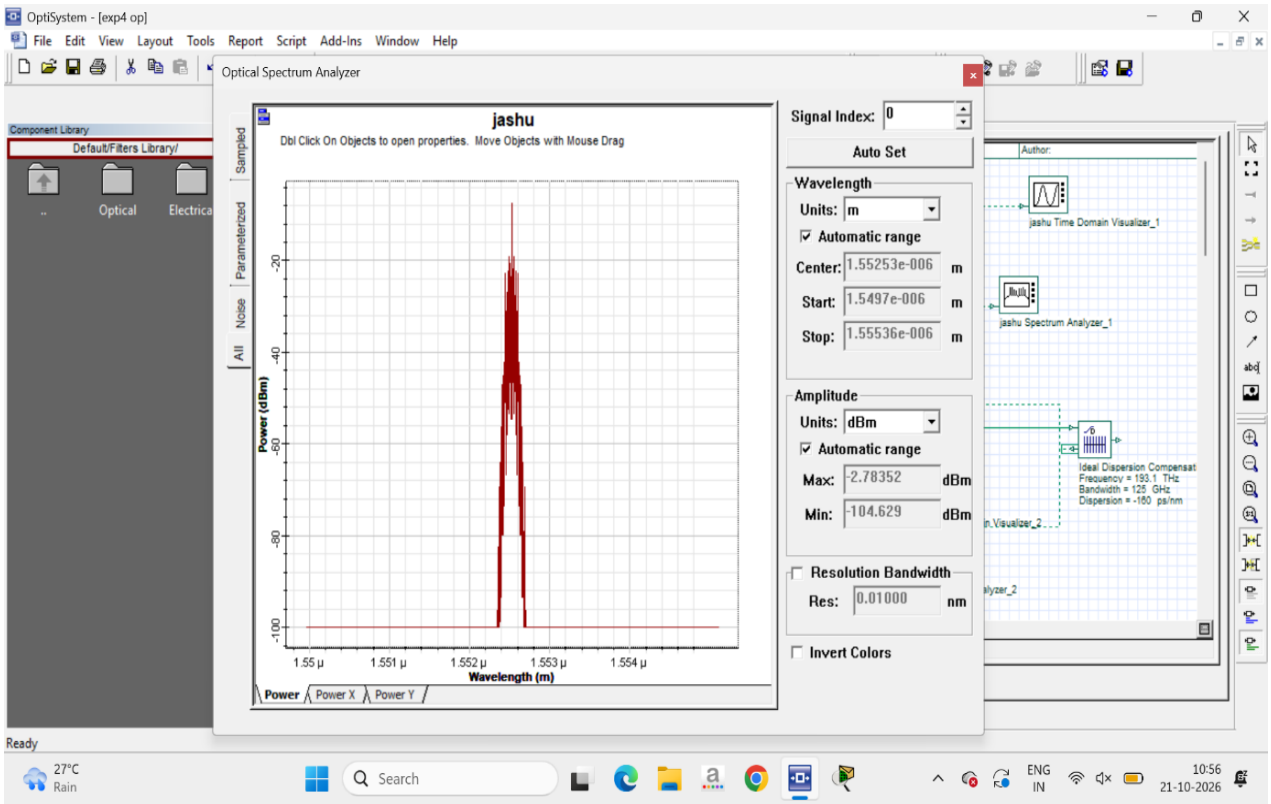
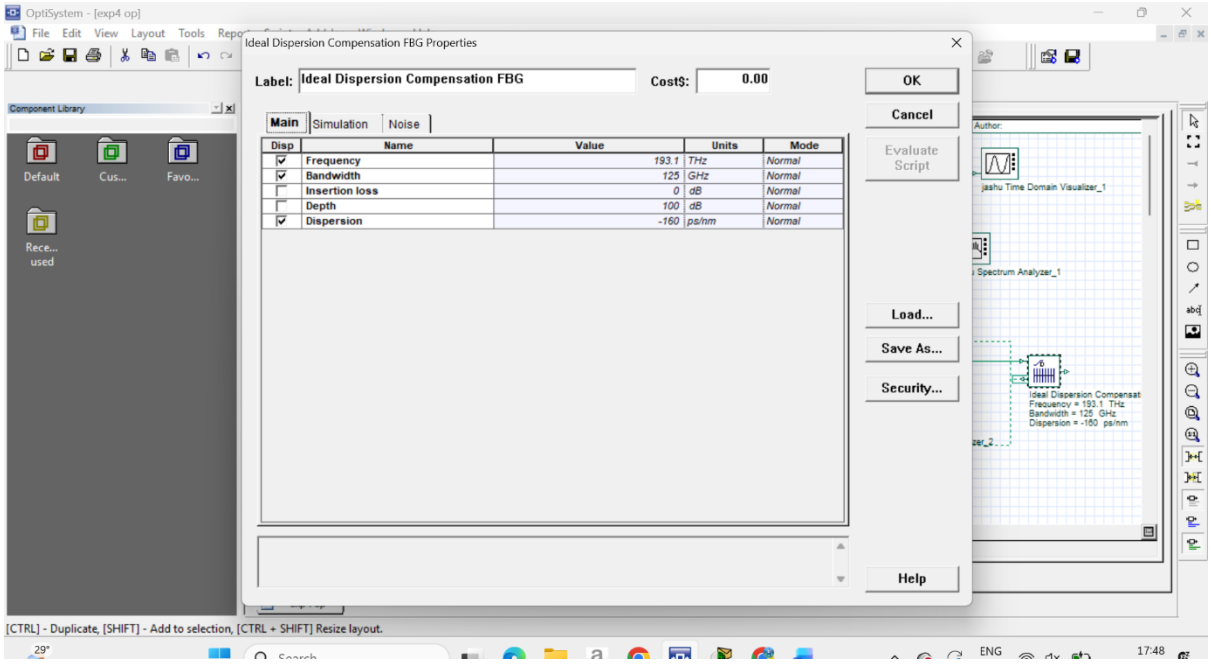
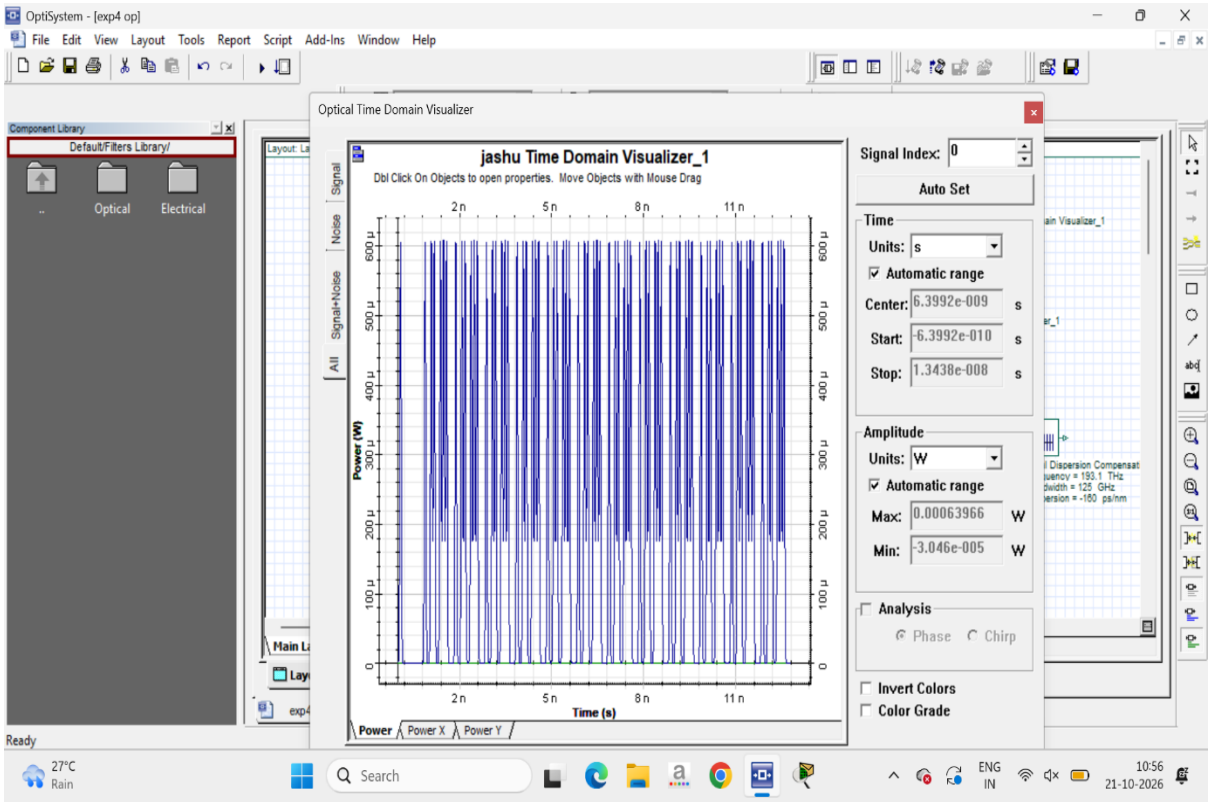

OptiSystem - [exp4 op]
File Edit View Layout Tools Report Script Add-Ins Window Help
Optical Spectrum Analyzer
jashu
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Signal Index: 0
Auto Set
Wavelength
Units: m
Automatic range
Center: 1.55253e-006 m
Start: 1.5497e-006 m
Stop: 1.55536e-006 m
Amplitude
Units: dBm
Automatic range
Max: -2.78352 dBm
Min: -104.629 dBm
Resolution Bandwidth
Res: 0.01000 nm
Invert Colors
Wavelength (m)
Power (dBm)
Power Power X Power Y
jashu Time Domain Visualizer_1
jashu Spectrum Analyzer_1
Ideal Dispersion Compensat
Frequency = 193.1 THz
Bandwidth = 125 GHz
Dispersion = -160 ps/nm
Component Library
Default/Filters Library/
Optical
Ready
27°C Rain
Search
ENG IN
10:56 21-10-2026

Figure 2: Gaussian pulse after 10 km propagation in SMF

A.



B.



C.

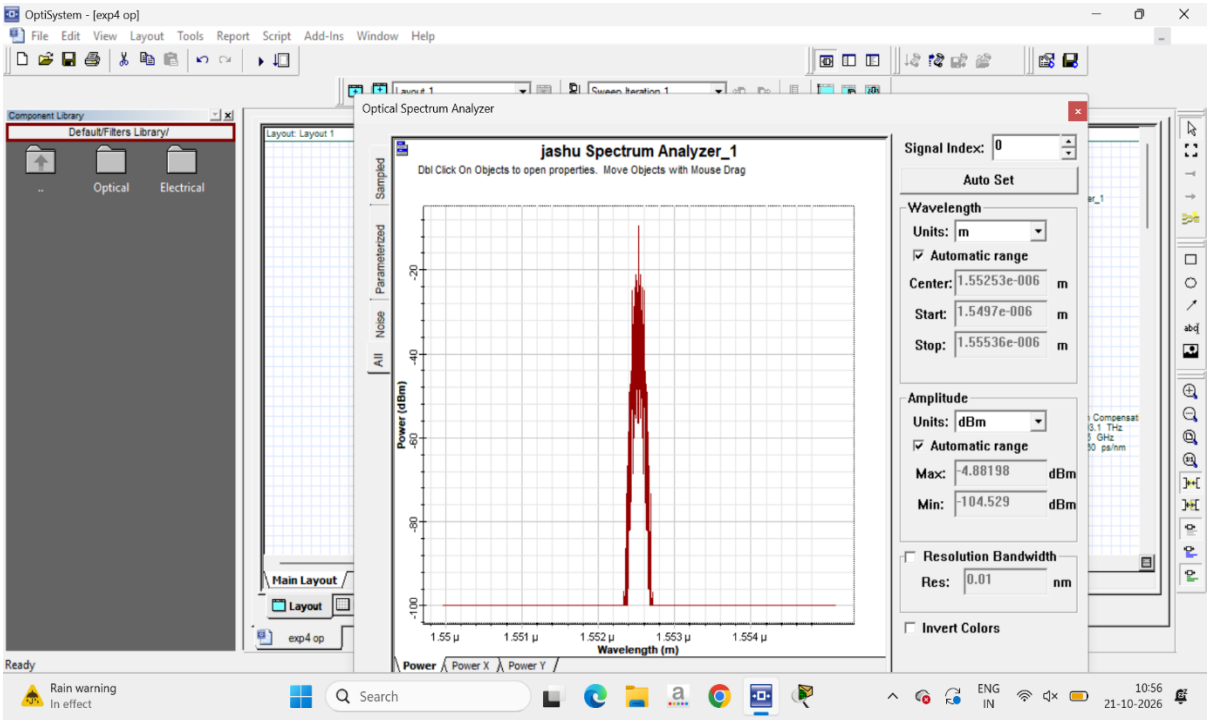


Figure 3: Parameters of Ideal Dispersion Compensation component

A.

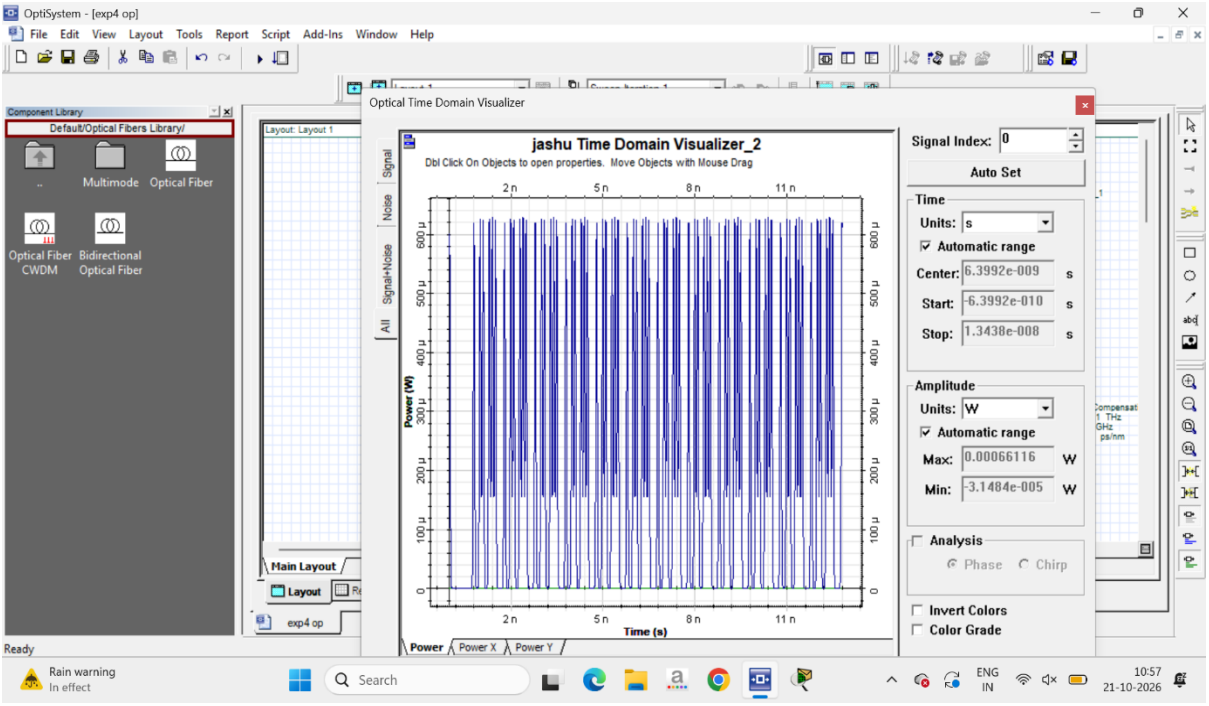
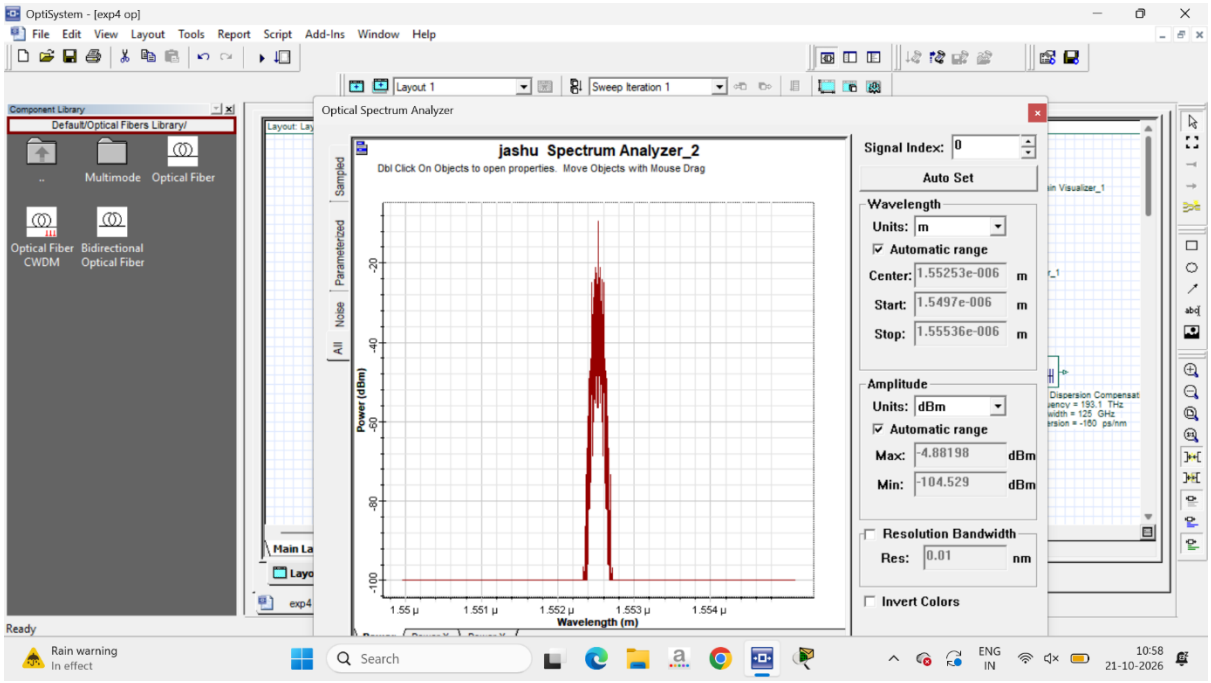


Figure 4: Pulse after dispersion compensation with Ideal Dispersion Compensation component

A.



8. OUTPUT PARAMETERS

Stage	Observation Point	Result	Figure
Initial pulse	Gaussian pulse generator output	Gaussian pulse, 12.5 ps	Figure 1 (A),(B)
After propagation	Output of 10 km SMF	Pulse width increased ≈ 4 times	Figure 2 (A),(B),(C)
Compensator setting	Ideal Dispersion Compensation FBG	Dispersion = –160 ps/nm	Figure 3 (A)
After compensation	Compensator output	Original pulse recovered (exact compensation)	Figure 4 (A)

Reflection

I learned how dispersion causes pulse spreading in optical fibers. I understood how dispersion compensation helps improve the quality of the received signal.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Design the OptiSystem layout	Project layout	Builds the complete layout with correct components and connections independently	Minor connection errors corrected	Needs guidance for components or connections	Unable to build the layout	20
Configure pulse, SMF and compensator parameters	Parameter configuration	Sets all parameters (40 Gb/s, 12.5 ps, 193.1 THz, 10 km, –160 ps/nm, 125 GHz) correctly	Minor parameter errors	Needs guidance for several parameters	Unable to configure parameters	20
Observe pulse broadening in SMF	Pulse observation	Captures and interprets initial and broadened pulses correctly	Minor interpretation errors	Limited observation / needs help	Unable to obtain or interpret pulses	20
Verify compensation using ideal dispersion component	Compensation verification	Verifies recovery of the pulse and explains residual effects of settings	Mostly correct verification	Partial verification with assistance	Unable to verify compensation	20
Document findings	Report and conclusion	Complete report with justified conclusion	Minor omissions	Incomplete record / weak conclusion	Report / conclusion inadequate	20

Result

Dispersion compensation was studied using OptiSystem simulation. The effect of dispersion on the transmitted optical signal was observed.